

Backend Integration Guide

LeadMaster AI — frontend chalane ke liye backend me kya integrate karna hai · 2026-08-01

Sabse pehle ye samajh lijiye: frontend chalane ke liye aapko koi bhi third-party API integrate karne ki **zarurat nahi hai**. App abhi bhi chalta hai, login hota hai, leads add hote hain, exports bante hain.

Third-party keys sirf **lead discovery** (automatic leads dhoondhna), **payments** aur **emails** ke liye chahiye. Inke bina wo features **fail nahi hote** — wo bas khaali result dete hain ya saaf error message dikhate hain. App kabhi bhi nakli data nahi banata.

Integration levels — ek nazar me

Level	Kya chahiye	Iske baad kya milega
Level 0 Zaroori	PostgreSQL + Redis JWT_SECRET_KEY	Poora app chalega: login, signup, manual leads, CSV import, website scanner, exports, dashboard, team, settings
Level 1 Lead discovery	GOOGLE_MAPS_API_KEY	Lead Search se automatic leads milenge, Map Search par pins dikhenge
Level 2 Better quality	MAPPLS_*, OPENAI_API_KEY	India ke leads behtar, AI-written lead summaries
Level 3 Production	SMTP_*, STRIPE_*, GOOGLE_CLIENT_*	Real emails, real payments, Google sign-in

Suggestion: Level 0 se shuru kijiye (already ho chuka hai). Phir sirf GOOGLE_MAPS_API_KEY add kijiye — usi ek key se app ka sabse bada feature (automatic lead search) zinda ho jaata hai.

Level 0 — Zaroori (bina iske app start hi nahi hoga)

Ye 3 cheezein chahiye

Cheez	Kyun	Aapke system par
PostgreSQL	Saara data yahin store hota hai	Ready — portable copy .devtools\pgsql me
Redis	OTP codes, rate limiting, Celery queue	Ready — dev substitute scripts/run_local_redis.py
JWT_SECRET_KEY	Login tokens sign karne ke liye	Set hai (dev value) — production me badalna zaroori

Production ke live nava JWT secret banavein

```
cd backend
venv\Scripts\python.exe -c "import secrets; print(secrets.token_urlsafe(64))"
```

Output ko backend/.env me daalein:

```
JWT_SECRET_KEY=<j0-output-aaya>
```

Ye badalte hi sabhi logged-in users logout ho jayenge (purane tokens invalid ho jaate hain). Isliye production launch se pehle hi set kar dijiye.

Level 0 par kya-kya poora kaam karta hai

- Signup, login, logout, password reset, OTP se login
- Manual lead add karna (Lead Database > Add Lead)
- **CSV import** — apni existing lead list upload karke import karna

- **Website Scanner** — kisi bhi website se email, phone, GSTIN, social links nikalna
- **Export Center** — CSV / Excel / PDF / JSON download
- Dashboard, charts, analytics, lead scoring (deterministic), deduplication
- Team invitations, settings, credit tracking

Yaani: agar aapke paas already leads ki list hai (Excel/CSV me), to aapko kisi API key ki zarurat hi nahi. Import kijiye aur poora app use kijiye.

Level 1 — Google Maps key (sabse zyada important)

Ye ek key 3 features ek saath zinda kar deti hai: Lead Search (automatic leads), Map Search (pins), aur address geocoding.

Kaise banayein — step by step

- 1 `console.cloud.google.com` par jaayein aur ek project banayein.
- 2 Left menu > **APIs & Services** > **Library** me jaakar ye **2 APIs enable** karein:
 - **Places API (New)** ← lead search ke liye
 - **Geocoding API** ← address se lat/lng ke liye
- 3 **APIs & Services** > **Credentials** > **Create Credentials** > **API key**.
- 4 Key copy karein.
- 5 **Billing enable karein** — Google Places bina billing account ke kaam nahi karta (free tier milta hai, par card add karna padta hai).

Kahan daalein

```
# backend/.env
GOOGLE_MAPS_API_KEY=AIZA...aapki-key
```

Phir backend restart karein.

Dhyan dein: ye **Places API (New)** maangta hai, purana "Places API" nahi. Agar purana wala enable kiya to har search 400 error dega. Google Cloud Library me dono alag-alag listed hain — naam me "(New)" hona chahiye.

Kaise verify karein ki chal gaya

- 1 App me login karein → **Lead Search** page kholein.
- 2 Search box me likhein: `switchgear manufacturers in Pune`
- 3 Search dabayein. Provider panel me **Google Places** ke saamne green tick aur lead count dikhna chahiye.
- 4 Agar 0 result aaye — provider panel me reason likha hoga (key missing / API not enabled / quota).

Bina key ke kya hota hai

Search chalti hai, **0 results** aate hain, aur provider panel batata hai ki "GOOGLE_MAPS_API_KEY is not configured". Koi credit nahi katta. App nakli leads nahi banata — ye jaan-boojh kar aisa design kiya gaya hai.

Level 2 — Quality badhane ke liye (optional)

Mappls (MapmyIndia) — India ke leads ke liye

India ke chhote businesses ka data Google se behtar hota hai. Isliye dono saath me chalane par zyada leads milte hain (duplicate automatically hat jaate hain).

- 1 `apis.mappls.com` par account banayein.
- 2 Console me ek project banakar **Client ID** aur **Client Secret** lein.

```
MAPPLS_CLIENT_ID=aapka-client-id
MAPPLS_CLIENT_SECRET=aapka-client-secret
```

OpenAI — AI lead summaries ke liye

Iske bina bhi lead scoring chalti hai (email/phone/GSTIN/website ke basis par real deterministic scoring). OpenAI add karne se sirf **summary text** zyada natural ho jaata hai.

```
OPENAI_API_KEY=sk-...
AI_SCORING_MODEL=gpt-4o-mini
```

Agar OpenAI down ho ya key galat ho, to search fail nahi hoti — wo automatically deterministic scoring par chali jaati hai.

Bing Search — ■■ shayad na mile

Microsoft ne **11 August 2025** ko standalone Bing Search API band kar diya. **Naye Azure accounts par ye ab nahi milta.** Agar aapke paas purana resource hai to chal jaayega, warna ise chhod dijiye — Company Website Search wahi kaam bina kisi key ke karta hai.

```
BING_SEARCH_API_KEY=aapki-key    # sirf purane Azure account par
```

Level 3 — Production ke liye

SMTP — real emails bhejne ke liye

Abhi emails **bhejte nahi, sirf backend console me print hote hain**. Development ke liye ye theek hai (verification link terminal me dikh jaata hai), par real users ke liye SMTP chahiye.

```
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=aapka@gmail.com
SMTP_PASSWORD=app-password
SMTP_USE_TLS=true
EMAIL_FROM=noreply@aapkadomain.com
```

Gmail ke liye normal password kaam nahi karega — Google Account > Security > **App passwords** se 16-character password banana padega (2FA on hona chahiye). Production ke liye SendGrid / AWS SES / Postmark behtar hain.

Stripe — payments ke liye

- 1 dashboard.stripe.com/test/apikeys se test keys lein.
- 2 Webhook banayein: Developers > Webhooks > Add endpoint →
https://aapka-domain/api/v1/billing/webhook
- 3 Signing secret copy karein.

```
STRIPE_SECRET_KEY=sk_test_...
STRIPE_PUBLISHABLE_KEY=pk_test_...
STRIPE_WEBHOOK_SECRET=whsec_...
```

Bina Stripe ke checkout button 400 error deta hai saaf message ke saath ("Stripe is not configured"). Baaki billing screens chalti rehti hain.

Google Sign-In

- 1 Google Cloud Console > Credentials > **OAuth client ID** > Web application.
- 2 **Authorized redirect URI** me exactly ye daalein (port aapke backend ka):
http://127.0.0.1:8001/api/v1/auth/google/callback

```
GOOGLE_CLIENT_ID=...apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=GOCSPX-...
GOOGLE_OAUTH_REDIRECT_URI=http://127.0.0.1:8001/api/v1/auth/google/callback
```

Zaroori: GOOGLE_OAUTH_REDIRECT_URI aur Google Console wali URI **bilkul same** honi chahiye — ek bhi character ka fark ho to Google redirect_uri_mismatch error dega.

Known limitation: Google se login karne par session **30 minute** me expire ho jaata hai. Backend ka OAuth callback sirf access token bhejta hai, refresh token nahi — isliye renew nahi ho paata. Email/password login me ye problem nahi (30 din chalta hai). Fix backend me karna hoga (callback me refresh token add karna).

Field encryption key

Sirf tab chahiye jab aap database me per-organization provider credentials store karna chahein. Aam use ke liye khaali chhod dijiye.

```
venv\Scripts\python.exe -c "from cryptography.fernet import Fernet; print(Fernet.generate_key().decode())"

PROVIDER_CREDENTIAL_ENCRYPTION_KEY=<output>
```

Ab use kaise karein — feature by feature

1. Pehli baar setup

- 1 App kholein → **Sign up** → naam, email, password, company name.
- 2 Onboarding wizard poora karein (industry, countries, team size). Ye aapke organization profile me save ho jaata hai.
- 3 Dashboard khul jayega — sab zero se shuru hoga, kyunki abhi koi lead nahi hai.

Naye account ko **Free plan: 100 credits** milte hain. Har search ~30 credits leti hai (5 providers × 5 results × cost), har website scan 1 credit.

2. Leads laane ke 3 tareeke

Tareeka	Kahan	Kya chahiye
Automatic search	Lead Search page	GOOGLE_MAPS_API_KEY (Level 1)
CSV import	Lead Database > Import CSV	Kuch nahi
Manual entry	Lead Database > Add Lead	Kuch nahi

CSV import kaise karein

Column headers flexible hain — Company, company_name, Organisation sab chalte hain. Sirf **company name column zaroori hai**, baaki optional.

```
Company, Industry, City, Email, Phone, GSTIN, Website
Apex Switchgear, Electrical, Pune, sales@apex.com, 9876543210, 27AAPFU0939F1ZV, apex.com
Nova Panels, Electrical, Thane, info@nova.in, 020-4567-8901, , nova.in
```

- Galat GSTIN/email/phone wali row **reject nahi hoti** — lead import ho jaata hai, bas wo ek field chhod diya jaata hai, aur aapko row number ke saath report milti hai.
- Duplicate leads automatically hat jaate hain (GSTIN → website → phone → naam+city ke basis par).
- Excel ka UTF-8 BOM, semicolon separator, sab handle ho jaata hai.

3. Website Scanner (koi key nahi chahiye)

Kisi bhi company ki website daaliye — email, phone, GSTIN (checksum verify hoke), social links, SSL, mobile-friendly, SEO score sab nikal aayega.

Private/internal address (localhost, 192.168.x.x) daalne par block ho jaayega — SSRF protection built-in hai.

4. Export

- 1 Export Center > **New Export**.
- 2 Source chunein (All / Filtered / Selected), format (CSV / Excel / PDF / JSON), columns.
- 3 File turant ban jaati hai aur Download Center me aa jaati hai.
 - **5,000+ rows** ka export background me jaata hai — uske liye Celery worker chalana padega.
 - Files **72 ghante** baad auto-delete ho jaati hain (Celery beat chal raha ho to).
 - Dashboard aur Analytics ki poori report bhi PDF/Excel me export ho sakti hai.

```
cd backend
venv\Scripts\celery.exe -A notifications.celery_app worker --pool=solo
venv\Scripts\celery.exe -A notifications.celery_app beat
```

Common problems aur unka fix

Problem	Wajah aur fix
Frontend par kuch load nahi ho raha / "Cannot reach the server"	Backend band hai, ya port match nahi kar raha. <code>.env.local</code> me <code>NEXT_PUBLIC_API_BASE_URL</code> aur backend ka <code>--port</code> same hone chahiye. Abhi dono 8001 hain.
Backend start hote hi <code>ConnectionRefusedError</code>	PostgreSQL band hai. <code>pg_ctl.exe -D .devtools\pgdata start</code> chalayein.
Login/signup par 500 error	Redis band hai (OTP aur rate limiting ke liye chahiye). <code>scripts/run_local_redis.py</code> chalayein.
Search se 0 results	Ye bug nahi hai . Koi provider configured nahi hai. Provider panel me har provider ke saamne reason likha hota hai. <code>GOOGLE_MAPS_API_KEY</code> add karein.
Search par 402 error	Credits khatam. Free plan me 100 credits hote hain. Billing se top-up karein, ya testing ke liye <code>CREDIT_METERING_ENABLED=false</code> set karein.
Google login par JSON error page	<code>GOOGLE_CLIENT_ID</code> set nahi hai. Ab app pehle check karta hai aur saaf toast dikhata hai bajaye error page par bhejne ke.
Google login ke 30 min baad logout	Known limitation — OAuth callback refresh token nahi bhejta. Email/password login use karein, ya backend callback me refresh token add karein.
Verification email nahi aaya	SMTP set nahi hai. Link backend ke console/terminal me print hota hai — wahan se copy karke browser me kholein.
Export download nahi ho raha	72 ghante purana export expire ho chuka hoga. Naya export banayein.
CORS error browser console me	<code>backend/.env</code> me <code>CORS_ORIGINS</code> me frontend ka URL hona chahiye (<code>http://localhost:3000</code>).

Sab kuch chal raha hai ya nahi — 60 second check

- 1 `http://127.0.0.1:8001/api/v1/health` kholein → `{"status":"ok"}` aana chahiye.
- 2 `http://127.0.0.1:8001/docs` kholein → Swagger UI me saare 96 endpoints dikhne chahiye.
- 3 `http://localhost:3000` par signup karein → dashboard khulna chahiye.
- 4 Lead Database > Add Lead se ek lead add karein → dashboard par "Total Leads 1" dikhna chahiye.
- 5 Export Center > New Export > CSV → file download hone chahiye jisme wo lead ho.

Ye 5 steps pass ho gaye to Level 0 poora kaam kar raha hai. Uske baad sirf API keys add karni hain jaisi zarurat ho.

Sabse chhota rasta: abhi jaise hai waise hi CSV import karke poora app use kar sakte hain. Automatic lead search chahiye to **sirf GOOGLE_MAPS_API_KEY** add kijiye — wahi ek key sabse zyada value deti hai.